

This is a medium-sized test document for performance benchmarking.

It contains multiple paragraphs discussing various aspects of Retrieval-Augmented Generation (RAG) systems.

Section 1: Introduction

Retrieval-Augmented Generation combines the strengths of information retrieval and text generation. Unlike standard language models that rely solely on parametric knowledge, RAG systems retrieve relevant information from an external knowledge base to generate more accurate and grounded responses.

Section 2: Architecture

A typical RAG system consists of four main components:

1. Document ingestion pipeline (PDF parsing, chunking, embedding)
2. Vector storage system (for efficient similarity search)
3. Retrieval mechanism (to find relevant context)
4. Generation component (LLM that produces grounded answers)

Section 3: Embedding Models

Embedding models convert text into numerical vectors that capture semantic meaning. Common choices include:

- OpenAI's text-embedding-3-small (384 dimensions)
- HuggingFace's all-MiniLM-L6-v2 (384 dimensions)
- Local sentence-transformers models

Section 4: Vector Stores

Vector stores enable efficient similarity search in high-dimensional spaces.

Popular options include:

- Qdrant (used in this project)
- FAISS
- Pinecone
- Weaviate

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